

Anjan Ranga Prasad

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SUMMARY

Master's Data Science student at Arizona State University with experience applying **Python, statistics, and machine learning** to histopathology, medical imaging, multi-omics, and unstructured data. Experienced in building **reproducible data and model-evaluation pipelines** for diagnostic analytics and healthcare AI research, with a focus on generating reliable insights that can support better patient care.

EDUCATION

Master of Science in Data Science, Analytics and Engineering

May 2027

Arizona State University, Tempe, AZ

3.83 GPA

Bachelor's Computer Science and Engineering

May 2025

PES University, Bangalore, India

TECHNICAL SKILLS

Programming / Data: Python, SQL, R, Pandas, NumPy, data acquisition, data cleaning, transformation, exploratory data analysis, JSON/CSV workflows

AI / Machine Learning: PyTorch, scikit-learn, ResNet, GraphSAGE, image classification, pattern recognition, feature engineering, algorithms, linear algebra, cross-validation, statistical analysis, model evaluation

Healthcare / Research Engineering: histopathology, medical imaging, multi-omics, clinical NLP, signal processing, Grad-CAM, Git/GitHub, Jupyter, Linux/UNIX, HPC, FastAPI, reproducible experimentation, literature review, documentation

PROFESSIONAL EXPERIENCE

Engineering Trainee, Vehere Technologies Private Limited, India

Jan 2025 – Mar 2025

- Developed a **Python/FastAPI pipeline** that converted unstructured **.wav** recordings into speaker-labeled transcripts and structured JSON using diarization and transcription
- Achieved approximately **80%–90% accuracy in speaker labeling and transcription** on evaluated samples by integrating **pyannote.audio** diarization with **Whisper Large-v3**
- Parallelized each file across **4 CPU cores** and standardized JSON/CSV outputs, enabling reproducible batch validation, archival, and downstream analysis

PROJECTS

PCam Histopathology Classification and Interpretability

- Trained a **ResNet-18 classifier** to detect metastatic tissue in histopathology patches, achieving **0.866 ± 0.011 accuracy** and **0.950 ± 0.005 ROC-AUC** across **5 runs**
- Evaluated model behavior using **Grad-CAM**, decision-threshold analysis, and error review to identify tissue regions and failure patterns influencing classification

MedMNIST Medical Imaging Benchmark Reproduction

- Benchmarked **ResNet-50** and **3D ResNet-50** through **90 training and evaluation runs** across **18 MedMNIST datasets** spanning 12 two-dimensional and 6 three-dimensional imaging tasks
- Identified dataset-level performance gaps by comparing mean AUC and standard deviation across **5 runs per dataset** against published baselines

TCGA-BRCA Multi-Omics Breast Cancer Classification

- Integrated RNA-expression and somatic-mutation data from **223 breast cancer patients** to build a **4,460-edge patient-similarity graph** for PAM50 subtype classification
- Validated a combined RNA-mutation **GraphSAGE model** on **216 patients across 4 PAM50 subtypes** using stratified **5-fold cross-validation**, achieving **0.898 ± 0.043 accuracy** and **0.905 ± 0.044 macro-F1**

PUBLICATION

YOLOv8,v9 and YOLOv10 Comparative Study for Autorickshaw and Number Plate Detection

- Published in Springer LNNS**; annotated a custom **5.5K-image dataset** and benchmarked **16 YOLO variants**, achieving **0.795 mAP@50** for autorickshaws and **0.745** for number plates